

SemEval 2026 task 12

Anonymous ACL submission

Abstract

1 Introduction

Recent large language models have achieved impressive capabilities across many NLP tasks. However, abductive reasoning, which involves inferring the most plausible cause of an observed outcome, remains difficult for current model, as it requires causal inference under incomplete or noisy evidence. This challenge becomes even more pronounced in real-world event understanding, where relevant evidence is often sparse, indirect or intermixed with irrelevant information.

In this work, we examine the task setting where a system is given a description of an observed event along with collection of retrieved documents that may include both relevant and distractor information. The goal is to determine plausible causes from a set of candidates. Solving this task requires the model to identify which evidences genuinely support a causal interpretation while disregarding irrelevant content, making it a targeted evaluation of abductive event reasoning.

[To be completed after finalizing the experimental setup]

2 Related Work

Early work on abductive reasoning has focused mainly on common sense reasoning and employed statistical NLP methods such as Pointwise Mutual Information (PMI), based on datasets such as COPA (Choice of Plausible Alternatives) and extensions that emphasize causal plausibility (Roemmele et al., 2011; Gordon et al., 2012). With the advent of large language models (LLMs), a growing body of work investigates the use of LLMs for causal inference tasks, where models directly select a cause from candidates given a context. The progression of research has shifted from causal reasoning to

causal inference, extending the scope from Causality for LLMs to LLMs for Causality (Ma, 2025a; Liu et al., 2025), aimed at solving complex formal inference tasks rather than model evaluation and enhancement.

Most existing LLM-based causal inference methods rely on prompting strategies such as in-context learning and chain-of-thought reasoning. Despite the high performance when the causal link is embedded in general world knowledge and linguistic patterns, prompting-based LLM lacks the logical reasoning required for intervention and counterfactual (Kıcıman et al., 2024). Moreover, it is prone to obscure true causal relations from associative patterns (Kıcıman et al., 2024; Jin et al., 2024) and often fail to follow normative causal inference rules like the “explaining away” effect (Dettki et al., 2025), where observing one sufficient cause reduces the probability of other independent causes.

Finetuning has been another widely examined method for improving the performance of LLMs on causal, abductive, and counterfactual reasoning tasks. As for complex tasks such as CORR2CAUSE (Jin et al., 2024) and causal graph validation (Susanti and Holsmoelle, 2025), finetuning yields clear improvements over general-purpose models. Finetuned Mistral-7B also exhibits improved performance on data-driven causal inference tasks (Cai et al., 2024). However, LLMs are still considered “causal parrots” that only succeed within the in-distribution data (Zečević et al., 2023), rather than possessing genuine structural causal reasoning. While fine-tuning can enhance performance, its capability typically lacks robustness and generalization (Jin et al., 2024).

To improve factuality and better process domain-specific queries, recent work explores integrating Retrieval-Augmented Generation (RAG) into LLM-based causal inference, by grounding the reasoning in external evidence (Lewis et al., 2021; Arslan et al., 2024). Iterative retrieval-based methods fur-

ther interleave reasoning and evidence acquisition, reporting improved robustness in multi-hop and distractor-rich settings (Yao et al., 2023; Trivedi et al., 2023). Beyond general knowledge grounding, hybrid approaches such as LLM Assisted Causal Recovery (LACR) (Zhang et al., 2024), which integrates RAG and constraint-based prompting into a causal graph discovery method, specifically retrieving cause-effect detected evidence to distinguish spurious correlations from true causes. However, RAG and extensions remain limited by the coverage and quality of the external knowledge base and the computational complexity (Gao et al., 2024; Rackauckas, 2024).

While LLMs have proven exceptional capability in reasoning benchmarks, as tested in (Ma, 2025b), one major disadvantage of them is that they are time consuming and expensive to run, especially in ranking or retrieving from a large collection. A possible solution for this is applying current advancement in information retrieval tasks (IR) as tested by in (Xiao et al., 2024). These approaches have the major advantage of being able to preprocess documents before inference time.

However, Xiao et al. also found that "current state-of-the-art retriever models may still be far from being competent for playing the role of assisting LLMs, especially in reasoning-intensive tasks". Therefore, further adjustment than just using retrievers is required. For example in (Zandie et al., 2023), a cross-encoder (Reimers and Gurevych, 2019) based on BERT (Devlin et al., 2019) is augmented by generated texts from a pretrained Bidirectional and Autoregressive Trans-former (BART) (Lewis et al., 2020) model.

3 Methodology

[Working progress, subjected to changes]

Given the current state of LLMs in abductive reasoning tasks, we aim to create a system that not only utilize LLMs common sense reasoning capabilities but also their abilities to assist in reasoning as retrieval tasks, as done in (Zandie et al., 2023). Our setup includes 2 parts, a base ranker that uses semantic vector representations and semantic textual similarity (STS), a re-ranker that uses RAG with a LLM.

Our base ranker utilize a LLM that to perform commonsense Abductive Natural Language Generation (α NLG). In this step multiple possible explanation for a possible real-world event is gen-

erated as texts. These texts are then compared to documents in our multiple choice test using a cross-encoder based on BERT and a aggregated STS is calculated for ranking.

Our re-ranker would then select a subset from the top-n documents from the previous ranking. Here, we utilize a LLM augmented with background knowledge documents to perform Abductive Natural Language Inference (α NLI) as a multiple Q&A task

References

- Muhammad Arslan, Hussam Ghanem, Saba Munawar, and Christophe Cruz. 2024. *A survey on rag with llms*. *Procedia Computer Science*, 246:3781–3790. 28th International Conference on Knowledge Based and Intelligent information and Engineering Systems (KES 2024).
- Hengrui Cai, Shengjie Liu, and Rui Song. 2024. *Is knowledge all large language models needed for causal reasoning?* *Preprint*, arXiv:2401.00139.
- Hanna M. Dettki, Brenden M. Lake, Charley M. Wu, and Bob Rehder. 2025. *Do large language models reason causally like us? even better?* *Preprint*, arXiv:2502.10215.
- Jacob Devlin, Ming-Wei Chang, Kenton Lee, and Kristina Toutanova. 2019. *Bert: Pre-training of deep bidirectional transformers for language understanding*. *Preprint*, arXiv:1810.04805.
- Yunfan Gao, Yun Xiong, Xinyu Gao, Kangxiang Jia, Jinliu Pan, Yuxi Bi, Yi Dai, Jiawei Sun, Meng Wang, and Haofen Wang. 2024. *Retrieval-augmented generation for large language models: A survey*. *Preprint*, arXiv:2312.10997.
- Andrew Gordon, Zornitsa Kozareva, and Melissa Roem-mele. 2012. *SemEval-2012 task 7: Choice of plausible alternatives: An evaluation of commonsense causal reasoning*. In **SEM 2012: The First Joint Conference on Lexical and Computational Semantics – Volume 1: Proceedings of the main conference and the shared task, and Volume 2: Proceedings of the Sixth International Workshop on Semantic Evaluation (SemEval 2012)*, pages 394–398, Montréal, Canada. Association for Computational Linguistics.
- Zhijing Jin, Jiarui Liu, Zhiheng Lyu, Spencer Poff, Mrinmaya Sachan, Rada Mihalcea, Mona Diab, and Bernhard Schölkopf. 2024. *Can large language models infer causation from correlation?* *Preprint*, arXiv:2306.05836.
- Emre Kıcıman, Robert Ness, Amit Sharma, and Chen-hao Tan. 2024. *Causal reasoning and large language models: Opening a new frontier for causality*. *Preprint*, arXiv:2305.00050.

182	Mike Lewis, Yinhan Liu, Naman Goyal, Marjan Ghazvininejad, Abdelrahman Mohamed, Omer Levy, Veselin Stoyanov, and Luke Zettlemoyer. 2020. BART: Denoising sequence-to-sequence pre-training for natural language generation, translation, and comprehension . In <i>Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics</i> , pages 7871–7880, Online. Association for Computational Linguistics.	236
183		237
184		238
185		239
186		240
187		241
188		242
189		
190		
191	Patrick Lewis, Ethan Perez, Aleksandra Piktus, Fabio Petroni, Vladimir Karpukhin, Naman Goyal, Heinrich Küttler, Mike Lewis, Wen tau Yih, Tim Rocktäschel, Sebastian Riedel, and Douwe Kiela. 2021. Retrieval-augmented generation for knowledge-intensive nlp tasks . <i>Preprint</i> , arXiv:2005.11401.	243
192		244
193		245
194		246
195		
196		
197	Xiaoyu Liu, Paiheng Xu, Junda Wu, Jiaxin Yuan, Yifan Yang, Yuhang Zhou, Fuxiao Liu, Tianrui Guan, Hao-liang Wang, Tong Yu, Julian McAuley, Wei Ai, and Furong Huang. 2025. Large language models and causal inference in collaboration: A survey . <i>Preprint</i> , arXiv:2403.09606.	247
198		248
199		249
200		250
201		
202		
203	Jing Ma. 2025a. Causal inference with large language model: A survey . <i>Preprint</i> , arXiv:2409.09822.	
204		
205	Jing Ma. 2025b. Causal inference with large language model: A survey . <i>Preprint</i> , arXiv:2409.09822.	
206		
207	Zackary Rackauckas. 2024. Rag-fusion: A new take on retrieval augmented generation . <i>International Journal on Natural Language Computing</i> , 13(1):37–47.	
208		
209		
210	Nils Reimers and Iryna Gurevych. 2019. Sentence-bert: Sentence embeddings using siamese bert-networks . <i>Preprint</i> , arXiv:1908.10084.	
211		
212		
213	Melissa Roemmele, Cosmin Bejan, and Andrew Gordon. 2011. Choice of plausible alternatives: An evaluation of commonsense causal reasoning . In <i>AAAI Spring Symposium - Technical Report</i> .	
214		
215		
216		
217	Yuni Susanti and Nina Holsmoelle. 2025. Prompting or fine-tuning? exploring large language models for causal graph validation . <i>Preprint</i> , arXiv:2406.16899.	
218		
219		
220	Harsh Trivedi, Niranjan Balasubramanian, Tushar Khot, and Ashish Sabharwal. 2023. Interleaving retrieval with chain-of-thought reasoning for knowledge-intensive multi-step questions . In <i>Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)</i> , pages 10014–10037. Association for Computational Linguistics.	
221		
222		
223		
224		
225		
226		
227		
228	Chenghao Xiao, G Thomas Hudson, and Noura Al Moubayed. 2024. Rar-b: Reasoning as retrieval benchmark . <i>Preprint</i> , arXiv:2404.06347.	
229		
230		
231	Shunyu Yao, Jeffrey Zhao, Dian Yu, Nan Du, Izhak Shafran, Karthik R Narasimhan, and Yuan Cao. 2023. React: Synergizing reasoning and acting in language models . In <i>The Eleventh International Conference on Learning Representations</i> .	
232		
233		
234		
235		